LPCN: Lightweight Precise Classification Network for Hyperspectral Remote Sensing Imagery Based on Multiobjective Optimization

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***Abstract*— Hyperspectral remote sensing image (HSI) has the unique advantages of spectral continuity as well as synchronous acquisition of both image and spectra of objects, which can achieve precise classification. For HSI classification, the deep learning (DL) methods have been fully developed due to its powerful layer-by-layer nonlinear feature learning ability, which is currently the mainstream method. However, current classification networks often refer directly to the fixed framework of natural image processing and usually only focus on accuracy as an optimization goal, resulting in poor HSI data adaptation capability and parameter redundancy. In addition, HSIs usually have dozens of types of ground objects, with similar and mixed spectra, and the phenomenon of overlapping clusters is aggravated, making it difficult to distinguish similar objects. In this article, a lightweight precise classification network (LPCN) for HSI based on multiobjective optimization was proposed. In LPCN, traditional fixed architectures are avoided, hierarchical lightweight search spaces are designed, and spectral attention mechanisms for similar objects are incorporated to improve separability. Moreover, the accuracy and parameter function of the classification network are independently modeled and optimized simultaneously for reducing the amount of network parameters. The effectiveness of LPCN is proved by experiments with three HSI datasets, with 16, 16, and 22 types of ground objects.**

***Index Terms*— Deep learning (DL), hyperspectral remote sensing, image classification, multiobjective optimization, neural architecture search.**

I. Introduction

REMOTE sensing Earth observation is an important means for large-scale and highly dynamic monitoring of surface elements and change information [1], [2]. By the end of 2022, there were 1177 in orbit remote sensing satellites worldwide. Different from multispectral or optical remote sensing images, hyperspectral remote sensing images (HSIs) have higher spectral resolution, with hundreds of spectral bands [3]. Thus, through the subtle differences between HSI spectral bands, it can finely distinguish ground objects [4]. Classification, as the foundation of HSI interpretation, has always been the main means of information acquisition and has played an important role in applications such as agricultural and forestry fine monitoring, landscape diversity, wetland mapping, national defense security, and disaster emergency response [5], [6], [7], [8], [9].

For HSI classification methods, they usually only used the rich spectral information or spatial characteristics in the early stage, such as spectral angle matching [10], support vector machine (SVM) [11], and hyperpixel segmentation method after dimensionality reduction [12]. However, due to the complex spatial distribution and spectral mixing of ground objects, there are spectral changes between similar ground objects, and only using spectral information or spatial features can easily lead to serious misclassification [13]. Thus, combining spatial structure information and spectral information is important for HSI classification.

The methods of combining spectrum and space were then proposed, and there are two main strategies. First, input spatial and spectral information of cascaded pixels, such as manual features of spatial texture and shape [14]. Second is the spatial context information modeling methods, such as conditional random fields (CRFs) [15], [16] and Markov random fields (MRFs) [17], [18]. Although the above methods have achieved certain classification performance, they rely on expert knowledge to design low-level manual features, and thus, they lack the ability to mine high-level information for the high-dimensional, complex, and nonlinear hyperspectral data.

With the rapid development of the field of artificial intelligence, deep learning (DL) technology, especially deep convolutional neural networks (DCNN), is currently the mainstream method for HSI classification due to its powerful layer-by-layer nonlinear feature learning ability and natural spatial–spectral information extraction ability [19]. First, similar to traditional model-driven research streams, spectral information, as an important feature of HSI, is primarily considered in the DL-based HSI classification networks. The

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input is a 1-D spectral vector, and the classification model only uses spectral information and representative methods such as the deep belief networks (DBNs) [20], stacked automatic encoders (SAEs) [21], and 1-D convolutional neural networks (1)-D CNNs) [22]. Obviously, such methods lose the spatial information possessed by remote sensing images, resulting in an increasingly obvious phenomenon of misclassification and island effects. In another branch, the combination of principal component analysis (PCA) and CNN is considered to reduce data redundancy and improve signal-to-noise ratio (SNR) [23], which can utilize the spatial information to a certain extent. The advantage lies in saving computational costs, but it is easy to cause the loss of spectral information for key distinguishing features.

In the traditional HSI classification method, it has been proven that integrating spatial information into spectral features can effectively improve the classification performance. The spatial block classification method based on the combination of spectral and spatial information was proposed, which fully utilizes the spectral information and neighborhood spatial information of HSIs. Among them, the joint extraction method of depth spectral–spatial information represented by 2-D CNN has received the most extensive research [24], [25], [26], [27], such as spatial–spectral residual network (SSRN) [28], spectral–spatial unified network (SSUN) [29], and spectral–spatial feature tokenization transformer (SSFTT) method [30]. Since the HSIs are generally regarded as a 3-D cube, the 3-D CNN methods [31], [32], [33] have been proposed to more effectively express the HSI data features, such as the 3-D spatial spectrum residual network (3)-D SSRN) [28], 3-D expansion convolution network (3)-D ECNet) [34], spectral–spatial-dependent global learning (SSDGL) framework [35], and hybrid network model based on 3-D CNN and scalable graph convolutional network [36].

Although sufficient spectral and spatial details have been incorporated into the deep classification network, effectively improving classification accuracy, when designing a deep network model manually, there is a high demand for the designer's network model design experience and knowledge in the relevant fields of HSIs. For different HSIs, the workload of manually designing a network architecture is excessive. Therefore, how to reduce the time and energy consumption of manual design and shift more work to computing devices is one of the current research trends in the design of DL-based HSI classification networks. A DCNN based on automatic search was first used for spectral–spatial classification of HSIs (1)-D/3-D auto CNN) [37], which addressed the above concerns to some extent. For the automatic search methods, there are three main contents [38], [39]: search space construction and search strategy design in the search phase, as well as the full training phase. Subsequently, other similar research methods were proposed, such as spectral feature perception evolving network (SFPEN) [40] and lightweight multiscale neural architecture search with spectral–spatial attention (LMSS-NAS) [41].

The above automatic search methods introduce the idea of automatically establishing a classification framework for HSI classification and achieve better performance. However, there are still significant gaps in search space construction and modeling optimization. First, in terms of data, the existing architecture search is mainly based on hyperspectral images for research, and there is no research on automated search of deep classification network architecture for high spatial resolution and hyperspectral resolution remote sensing images (H2 image). Second, in terms of method design, the higher spatial resolution of H2 images makes different subcategories of land objects diverse in scale and often with significant differences in scale, such as rapeseed planted on several acres and cabbage planted on several square meters. The fixed structure of spatial–spectral attention networks is difficult to adaptively capture the spatial contextual information of subcategories of land objects with diverse scales. Moreover, the current network design process has not taken into account the computational complexity of the model, which limits the practicality of hyperspectral classification models.

By analyzing the above issues, a more advantageous network architecture search process can be modeled as a multiobjective optimization problem (MOOP) that simultaneously optimizes the accuracy and parameters of the HSI classification network. Fortunately, the multiobjective evolutionary algorithm (MOEA) can provide a unified framework [42], [43], and it can achieve the global optimization search in the predefined classification network architecture search space. Furthermore, MOEAs have been successfully applied to HSI denoising [44], unmixing [45], [46], band selection [47], [48], [49], and clustering [50], [51]. Therefore, a lightweight precise classification network (LPCN) for HSI based on multiobjective optimization was proposed in this article. The contributions are summarized as follows.

1) *ALightweight Multiobjective Evolutionary Automatic Search Framework for HSI Precise Classification:* In the proposed LPCN, avoiding the traditional HSI fixed classification architecture, the automatic search mechanism based on the MOEA has been introduced. Moreover, the flexible and variable HSI precise classification network architecture was optimized from the lightweight and multiscale search space based on the evolutionary algorithm. In addition, the HSI ground objects classification accuracy and the amount of the network parameter have been modeled and optimized simultaneously based on the multiobjective optimization theory.
2) *Lightweight and Multiscale Search Space for HSI Classification Network:* The proposed LPCN method utilizes the diversified encoding combination optimization ability of evolutionary algorithms to construct a multiscale and lightweight search space with variable scales of computational operators such as convolution kernels and dilated convolution kernels. By optimizing the combination of different convolution kernel sizes, the spatial–spectral context information of different scale subclasses of land features is adaptively captured in the HSI, especially for H2 image.
3) *Two-Step Multiobjective Optimization Modeling Strategy for Hyperspectral Classification Architecture:* First,

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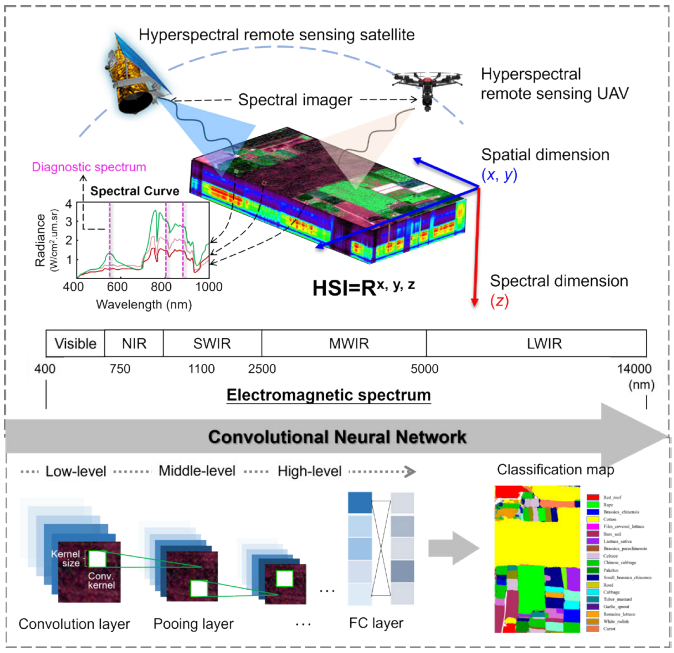


Fig. 1. Hyperspectral imaging and classification based on CNN.

through multiobjective modeling of the accuracy function and network parameter calculation function of the HSI classification network, achieve the accuracy growth under the tradeoff from the lightweight network. Furthermore, an accuracy growth function is embedded to construct a two-step dual objective function, avoiding the elimination of slightly larger models with better performance and obtaining superior practical models.

Furthermore, Section II presents the theory of HSI classification based on blocking mechanism. Section III describes the proposed LPCN framework. The experimental results and analyses are provided in Section IV. Finally, the conclusion and discussion are provided in Section V.

II. Related Research Background

A. Hyperspectral Remote Sensing Image Classification Based on CNN

As shown in Fig. 1, HSI is spectral continuous image data obtained by using space satellites and airborne platforms with spectral imager in the visible, near-infrared (NIR), shortwave infrared (SWIR), medium-wave infrared (MWIR), and long-wave infrared (LWIR) range of electromagnetic spectrum [3], [52]. In addition, the subtle spectral differences can be extended to diagnostic spectra, enabling differentiation between various ground features.

Since Chen et al. [53] first introduced DL methods into the field of HSI classification, CNNs based on block mechanism have been a hot topic in HSI classification, as shown in Fig. 1. The fully connected layer generally maps the hierarchical features learned by the network to the sample label space at the end of the CNN, as shown in the following equation:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right) \tag{1}$$

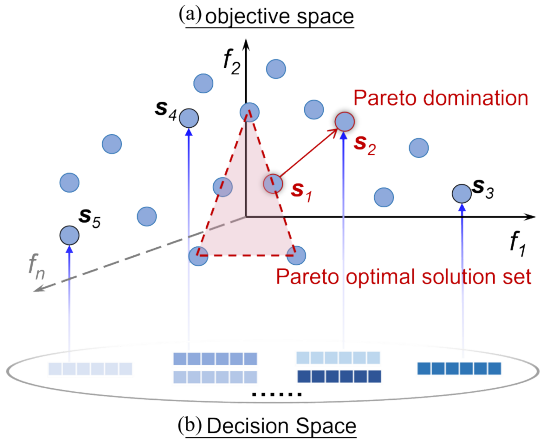


Fig. 2. Illustration for multiobjective optimization problem.

where w_i and b are the weights and offsets of neuronal connections, respectively. $f(\cdot)$ is the activation function, and common activation functions include Sigmoid, Tanh, and ReLU.

B. Multiobjective Optimization Problem

In many real-world applications, multiobjective optimization problems (MOOPs) are widely present [54]. The research object of this study is also a typical MOOP, which expects the best performance between accuracy and parameter quantity for the HSI classification network. In addition, the MOOP is presented, which has m decision variables and n target variables, as shown in the following equation:

$$\min \mathbf{y} = \mathbf{F}(\mathbf{s}) = (f_1(\mathbf{s}), f_2(\mathbf{s}), \ldots, f_n(\mathbf{s}))^T. \tag{2}$$

As shown in Fig. 2, assuming that $\mathbf{s}_1$ and $\mathbf{s}_2$ are two feasible solutions, and compared with $\mathbf{s}_1, \mathbf{s}_2$ is Pareto dominant if and only if it satisfies condition (3). Moreover, when $\mathbf{s}^\dagger \in \mathbf{S}$, all the Pareto optimal solutions make up the Pareto optimal solution set $PS^\dagger$, as shown in (4), $\succ$ denotes the Pareto dominant

$$\forall i = 1, 2, \ldots, n,\ f_i(\mathbf{s}_1) \le f_i(\mathbf{s}_2) \wedge$$
$$\exists j = 1, 2, \ldots, n,\ f_j(\mathbf{s}_1) < f_j(\mathbf{s}_2) \tag{3}$$

$$PS^\dagger \left\{\mathbf{s}^\dagger | \neg \exists \mathbf{s} \in \mathbf{S} : \mathbf{s} \succ \mathbf{s}^\dagger\right\}. \tag{4}$$

III. Proposed LPCN

The overall framework of the proposed LPCN method is shown in Fig. 3, and the detailed process is presented as follows.

A. Lightweight Search Space Embedded in HSI Spectral Attention Mechanism

In the proposed LPCN method, for obtaining a lightweight HSI classification model with low complexity and suitable for offline practical reasoning applications, the eight different types of lightweight computing operators form the operating modes within the module, which contains two pooling operations, four convolution operations, skip connections, and none, and are encoded as 0–7. Furthermore, with the continuous improvement of spatial resolution in hyperspectral images,

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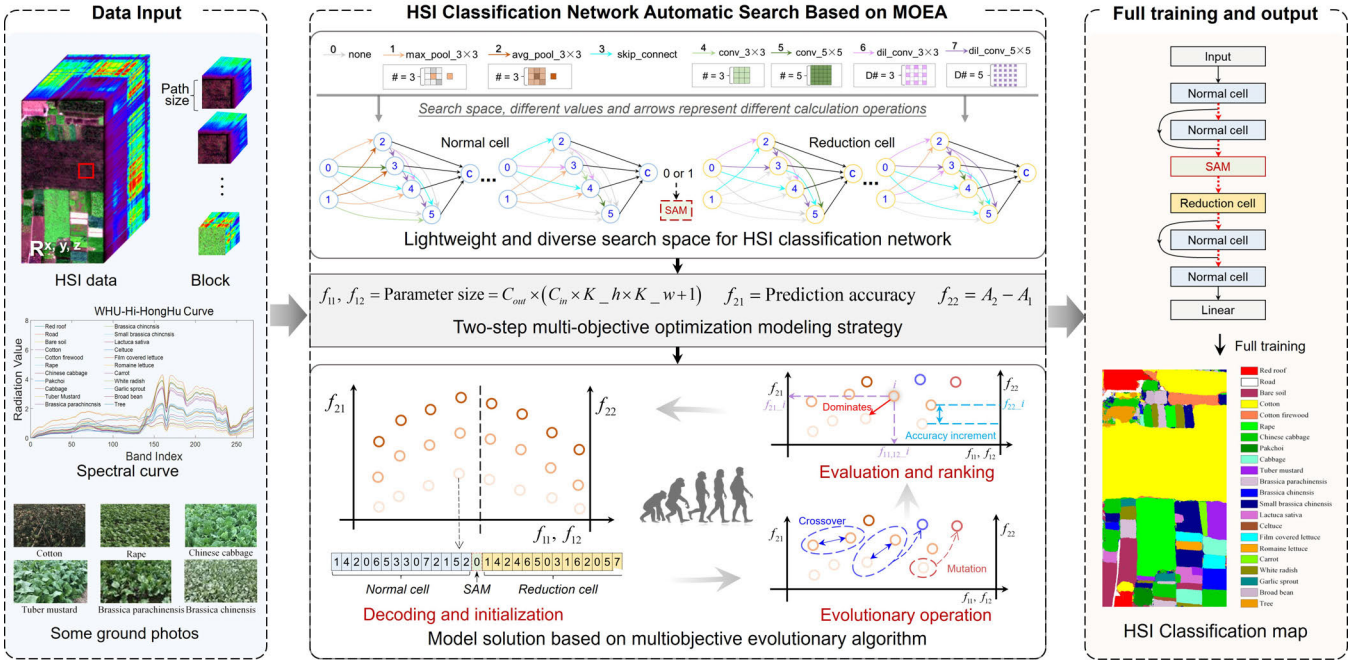


Fig. 3. Overall framework of the proposed LPCN method.

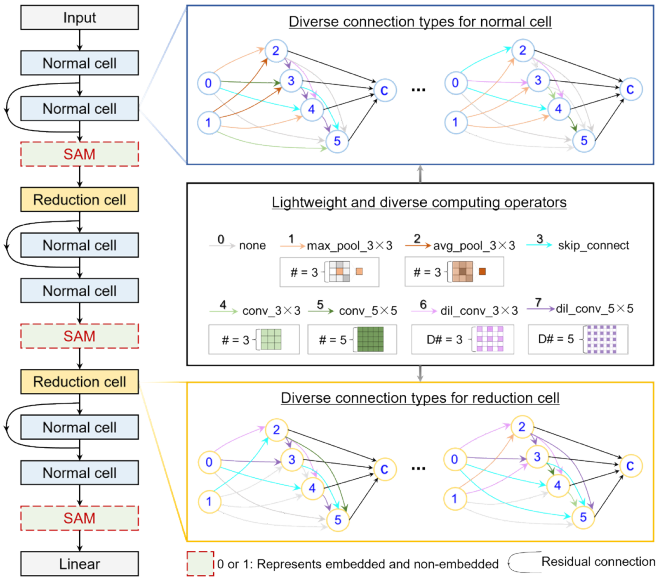


Fig. 4. Diversified search space for normal and reduction cells. ($n = 4$, number of reduction cells = 2, and the step of reduction cell = 2).

the scales of different subclasses of land use have become diverse, such as large-scale rapeseed and small-scale cabbage. The operator contains convolution kernels of different scales, which can adaptively meet the requirements of fine-grained classification of land use at different scales in this study. In addition, in order to better achieve the differentiation and precise classification of similar objects in HSI scenes, the spectral attention module (SAM) is introduced into the search space to achieve fine differentiation of similar ground objects on spectral segments, with 0 representing embedding and 1 representing nonembedding, as shown in Fig. 4.

Furthermore, the normal and reduction cells are designed to accommodate features of different scales in the HSI scenes. There are seven nodes in a normal or reduction cell, where © denotes the "Concatenate" function. In addition, the first two calculation nodes and ① are used as inputs for the next n nodes ②,③,④,⑤, . . . , which can fully learn information between multiple levels and avoid the problem of gradient vanishing. Moreover, these nodes are used to form the HSI deep classification network architecture search space by selecting different connection modes, and the SAM is embedded after the normal cell for selection. Therefore, the complete search space size Π for HSI classification network architecture evolutionary automatic search is shown in the following equation:

$$\begin{aligned}\Pi &= \underbrace{\left(n \times 2 + \frac{n \times (n-1)}{2}\right) \times 8 \times 2}_{\mathbf{NS}:\text{Noraml cell+SAM}} \\ &\quad \times \underbrace{\left(n \times 2 + \frac{n \times (n-1)}{2}\right) \times 8}_{\mathbf{R}:\text{Reduction cell}} \\ &= \left(n \times 2 + \frac{n \times (n-1)}{2}\right)^2 \times 128. \end{aligned} \tag{5}$$

B. Two-Step Multiobjective Optimization Modeling Strategy

First, in the evaluation of existing HSI classification networks, classification accuracy is usually only used as an evaluation indicator, leading to a continuous increase in classification network parameters and an exponential increase in computational complexity. However, computing resources at the edge of satellite, unmanned aerial vehicles (UAVs) and other devices are often limited, resulting in a relatively low level of practical application of the existing HSI classification network architecture. Thus, the multiobjective evaluation functions $\mathbf{F}^1$ for the HSI classification network parameter

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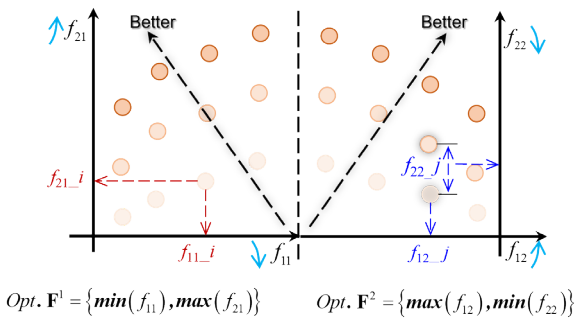


Fig. 5. Graphical two-step multiobjective modeling strategy. (f_{11}_i and f_{21}_i denote the $\mathbf{F}^1$ objective function values of the ith individual, and f_{12}_j and f_{22}_j represent the $\mathbf{F}^2$ objective function values of the jth individual).

quantity f_{11} and prediction accuracy f_{21} are first designed in the proposed LPCN method, as shown in (6).

In addition, considering the existing evolutionary optimization search process, HSI classification network individuals with slightly larger parameter quantities are usually immediately eliminated. However, due to insufficient training in the search stage, the accuracy performance after full training cannot be foreseen. This HSI classification network with slightly larger parameter quantity has the possibility of higher accuracy and is more selective in comparison. Therefore, another set of multiobjective functions $\mathbf{F}^2$ of HSI classification network parameter quantity f_{12} and accuracy increment f_{22} was further designed, as shown in (7) and Fig. 5

$$\begin{aligned} \text{Opt.}\mathbf{F}^1 &= \{\min(f_{11}), \max(f_{21})\} \\ &\begin{cases} f_{11} = C_{\text{out}} \times (C_{\text{in}} \times K_h \times K_w + 1) \\ f_{21} = \dfrac{N_c}{N_a} \end{cases} \end{aligned} \tag{6}$$

$$\begin{aligned} \text{Opt.}\mathbf{F}^2 &= \{\max(f_{12}), \min(f_{22})\} \\ &\begin{cases} f_{12} = C_{\text{out}} \times (C_{\text{in}} \times K_h \times K_w + 1) \\ f_{22} = A_2 - A_1 \end{cases} \end{aligned} \tag{7}$$

where C_{out} and C_{in} represent the number of output and input channels in the convolutional layer, respectively; and K_h and K_w represent the height and width of the convolutional kernel, respectively. N_a denotes all tested HSI classification samples, and N_c is the number of correctly predicted samples. A_2 and A_1 represent the HSI classification network prediction accuracy of individual solutions in two generations.

C. HSI Classification Model Solution Based on Multiobjective Evolutionary Algorithm

Based on the above designed lightweight search space and two-step dual multiobjective optimization functions, the multiobjective optimization problem of HSI classification can then be resolved by MOEA in this article, as shown in Fig. 3 and described in the following.

1) HSI Classification Network Individual Encoding and Population Initialization: In this article, the proposed LPCN method adopts an automatic search mechanism based on evolutionary algorithm, which can initialize multiple individual solutions of DL-based HSI classification network architecture

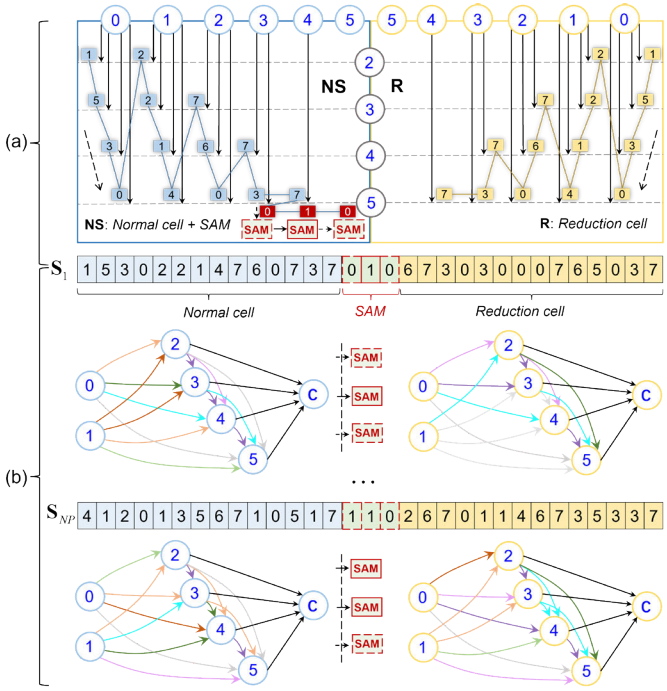


Fig. 6. Encoding and initialization. (a) Encoding of individual solution variables for HSI classification network architecture. (b) Population initialization for HSI classification network architecture. ($n = 4$ and number of reduction cells $= 2$).

at the same time to form a population-based global optimization search. In MOEAs, the individual represents the solution in the optimization problem, and the solution individuals in the proposed LPCN method represent different HSI classification network architectures and are effectively and flexibly encoded for subsequent iterative optimization, as shown in Fig. 6.

Specifically, in Fig. 6, the normal and reduction cells are first considered in the individual solutions, and there are eight different types of lightweight computing operators to choose from. Moreover, the SAM is embedded in the part of normal cell, which can achieve adaptive weight differentiation of similar features in the HSI scenes. Furthermore, the first two calculation nodes and ① are used as inputs for the next nodes ②,③,④, and ⑤, and the nodes before a certain node serve as input, such as there are three connection points, ① , and ② in front of node ③. Therefore, the individual solution $\mathbf{S}_1$ was encoded as "1-5-3-0-2-2-1-4-7-6-0-7-3-7-0-1-0-6-7-3-0-3-0-0-0-7-6-5-0-3-7," and there are NP HSI classification network architectures have been initialized simultaneously in a population.

2) Evolutionary Operation for HSI Classification Networks Reproduction: In the proposed LPCN method, in order to expand the diversity of the current population and regenerate more individual solutions of HSI DL classification network under the comparison of two-step multiobjective evaluation functions so that subsequent HSI classification network architectures can continue to effectively evolve, the crossover and mutation evolution operations in genetic algorithms are utilized in the proposed LPCN method until a better individual solution of the HSI classification network structure is obtained, as shown in Fig. 7.

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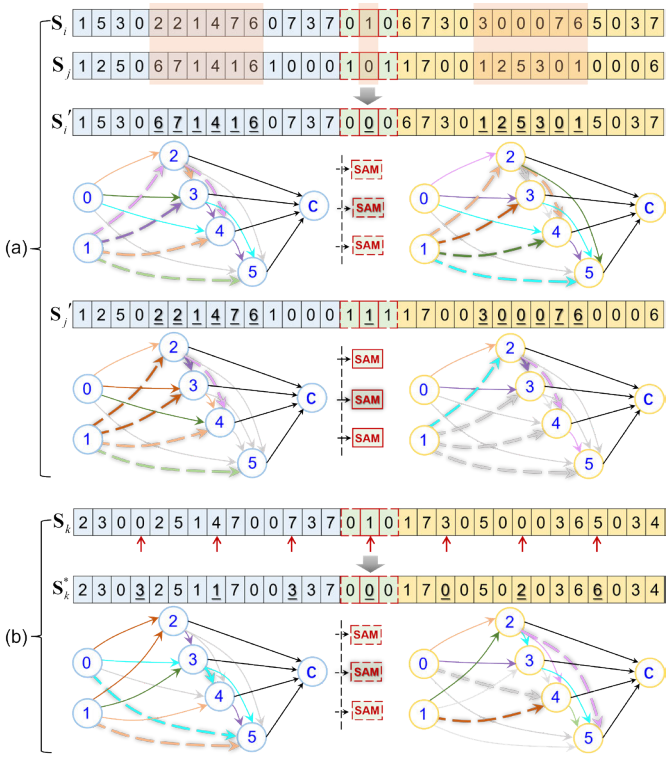


Fig. 7. Evolutionary operations. (a) Crossover. (b) Mutation. ($n = 4$ and number of reduction cells $= 2$).

First, three individual solutions $(\mathbf{S}_i, \mathbf{S}_j, \mathbf{S}_k)$ for HSI classification network architectures were selected, representing the i, j, and k individuals in one generation, respectively. The crossover operation was conducted between the individuals $\mathbf{S}_i$ = 1-5-3-0-2-2-1-4-7-6-0-7-3-7-0-1-0-6-7-3-0-3-0-0-0-7-6-5-0-3-7 and $\mathbf{S}_j$ = 1-2-5-0-6-7-1-4-1-6-1-0-0-0-1-0-1-1-7-0-0-1-2-5-3-0-1-0-0-0-6. In addition, there are six crossover points, and two new HSI classification networks have then been generated, which are $\mathbf{S}_i'$ = 1-5-3-0-6-7-1-4-1-6-0-7-3-7-0-0-0-6-7-3-0-1-2-5-3-0-1-0-0-0-6 and $\mathbf{S}_j'$ = 1-2-5-0-2-2-1-4-7-6-1-0-0-0-1-1-1-1-7-0-0-3-0-0-0-7-6-0-0-0-6. Moreover, the mutation operation was then conducted in the individual solution $\mathbf{S}_k$ = 2-3-0-0-2-5-1-4-7-0-0-7-3-7-0-1-0-1-7-3-0-5-0-0-0-3-6-5-0-3-4, there are seven mutation points, and the new solution $\mathbf{S}_k^*$ = 2-3-0-3-2-5-1-1-7-0-0-3-3-7-0-0-0-1-7-0-0-5-0-2-0-3-6-6-0-3-4 has been reproduced. Therefore, during the search process, the architecture of the HSI classification network is diverse, and through these two operations, the network structure can be flexibly changed to adapt to the hierarchical information of HSIs.

3) Evaluation and Ranking: After the evolutionary reproduction, there are 2 × NP individual solutions of the HSI classification networks. In order to keep a constant size for the population, the evaluation and ranking steps have been conducted in this section. As shown in Fig. 8, the dual multiobjective evaluation functions can be calculated by (6) and (7). Moreover, a super network for HSI classification was constructed in the proposed LPCN method, which has weight parameters on all node connections, and the weight parameters were updated with the evolutionary process. Therefore, the weight parameters of each searched HSI classification network

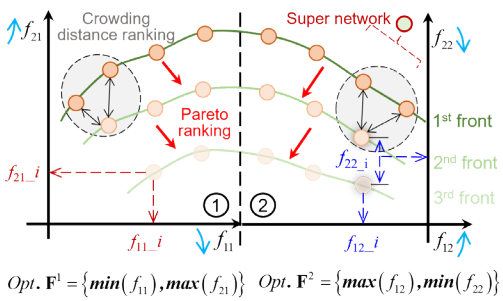


Fig. 8. Evaluation and ranking.

can obtain the pretrained weight parameters from the super network, which can effectively avoid the time-consuming issue of training from scratch.

Furthermore, when performing a two-step multiobjective ranking corresponding to the two-step dual multiobjective functions, first, the multiobjective nondominated ranking of the HSI classification network parameter quantity f_{11} and prediction accuracy f_{21} was conducted to obtain the sorting results on the left side. The multiobjective nondominated sorting of HSI classification network parameter quantity f_{12} and accuracy increment f_{22} was then performed to obtain the sorting results on the right side, and the Pareto fronts were finally obtained. In addition, when the number of individual solutions on the first Pareto front is less than or greater than the established population size, the crowding distance ranking is used to increase the diversity of the population to a certain extent, taking $\mathbf{F}^1$ as an example, as shown in (8). $f_{11}^{\min}$, $f_{11}^{\max}$, $f_{21}^{\min}$, and $f_{21}^{\max}$ represent the minimum and maximum values of the two objective functions in a population

$$\text{Crow_dist}_k = \frac{f_{11}(\mathbf{S}_{k-1}) - f_{11}(\mathbf{S}_{k+1})}{f_{11}^{\max} - f_{11}^{\min}} + \frac{f_{21}(\mathbf{S}_{k-1}) - f_{21}(\mathbf{S}_{k+1})}{f_{21}^{\max} - f_{21}^{\min}}. \tag{8}$$

Repeat the above evolutionary operations, evaluation, and ranking steps until the maximum number of iterations is reached. A competitive HSI classification network architecture in the Pareto optimal solution set was finally obtained, and the network individual with the highest accuracy was selected for full training, and output classification map and accuracy.

IV. Experiments and Analyses

In order to evaluate the effectiveness of the proposed approach, it was compared with traditional methods such as SVM, spatial–spectral information jointed SVM (S^2SVM), typical DL methods for HSI classification, such as SSRN [28] and SSUN [29], and the SSFTT method [30] and LMSS-NAS [41].

A. Experimental Data and Parameter Settings

Three HSI scene precise classification datasets were utilized in this article, which are the Indian Pines AVIRIS dataset [55], and the WHU-Hi-Hanchuan and WHU-Hi-Honghu datasets shared by the Intelligent Data Extraction and Analysis of Remote Sensing (RSIDEA) Group of Wuhan University [56]. The specific situation is shown in Fig. 9 and Table I.

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TABLE I
SPECIFIC SITUATION OF THE THREE HSI SCENE DATASETS

Indian Pines AVIRIS			WHU-Hi-HanChuan			WHU-Hi-HongHu		
Number of spectral bands		220	Number of spectral bands		270	Number of spectral bands		270
Spatial resolution		20m	Spatial resolution		0.109m	Spatial resolution		0.043m
Size		145×145	Size		1217×303	Size		940×475
#	Class	Samples size	#	Class	Samples size	#	Class	Samples size
1	Alfalfa	46	1	Strawberry	44735	1	Red_roof	14041
2	Corn-no-till	1428	2	Cowpea	22753	2	Road	3512
3	Grass-pasture	483	3	Soybean	10287	3	Bare_soil	21821
4	Grass-trees	730	4	Sorghum	5353	4	Cotton	163285
5	Corn-min-till	830	5	Water_spinach	1200	5	Cotton_firewood	6218
6	Corn	237	6	Water_melon	4533	6	Rape	44557
7	Grass-pasture-mowed	28	7	Greens	5903	7	Chinese_cabbage	24103
8	Hay-windrowed	478	8	Trees	17978	8	Pakchoi	4054
9	Oats	20	9	Grass	9469	9	Cabbage	10819
10	Soybean-no-till	972	10	Red_roof	10516	10	Tuber mustard	12394
11	Wheat	205	11	Grey_roof	16911	11	Brassica parachinensis	11015
12	Woods	1265	12	Plastic	3679	12	Brassica chinensis	8954
13	Soybean-min-till	2455	13	Bare_soil	9116	13	Small_brassica_chinensi	22507
14	Soybean-clean	593	14	Road	18560	14	Lactuca sativa	7356
15	Building-grass-tree-drive	386	15	Bright_objective	1136	15	Celtuce	1002
16	Stone-steel-towers	93	16	Water	75401	16	Film_covered_lettuce	7262
						17	Romaine_lettuce	3010
						18	Carrot	3217
						19	White_radish	8712
						20	Garlic_sprout	3486
						21	Broad_bean	1328
						22	Tree	4040

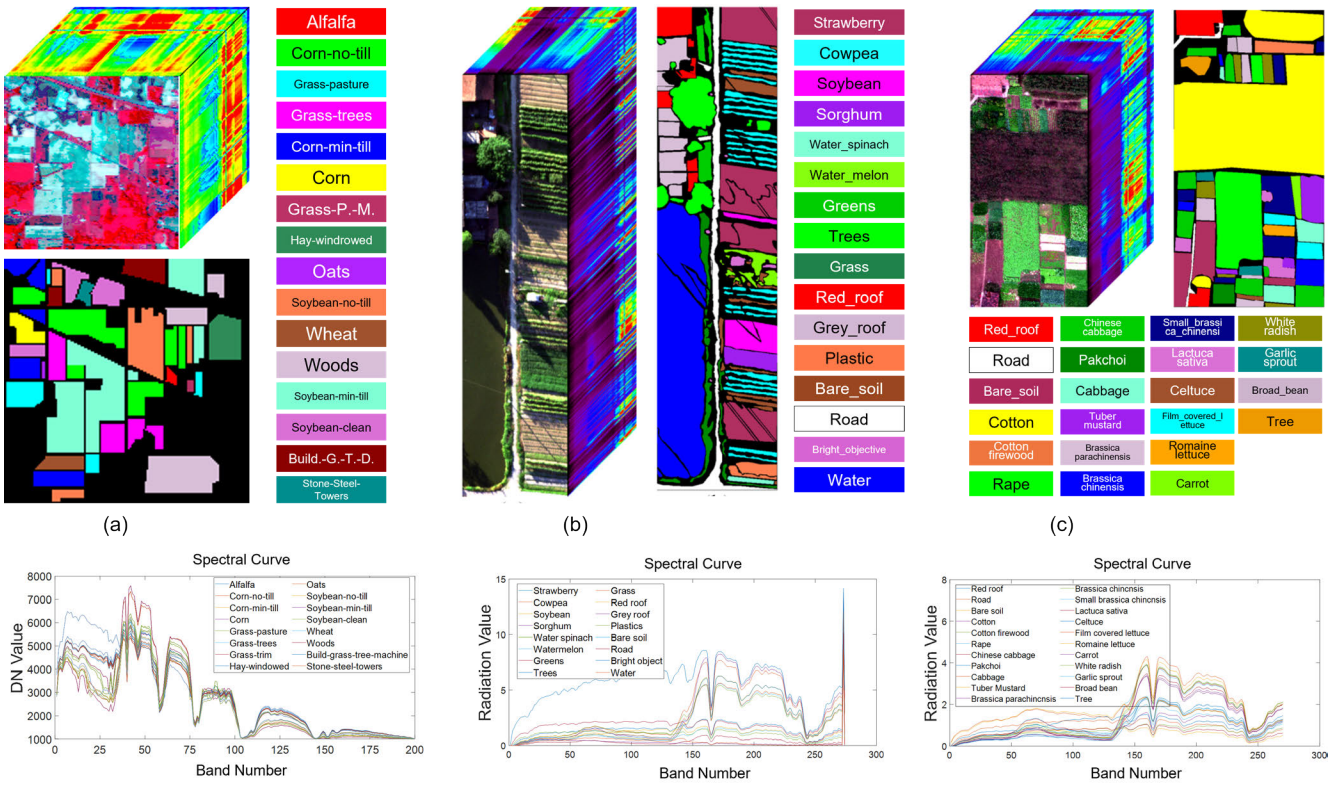


Fig. 9. Three HSI scene precise classification datasets. (a) Indian Pines AVIRIS. (b) WHU-Hi-Hanchuan. (c) WHU-Hi-Honghu.

The proposed LPCN method includes three stages: input, automatic search, and full training and output. The relevant experimental parameter settings for the three tested HSIs are shown in Table II. It can be found that in the search

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TABLE II
PARAMETER SETTINGS FOR THE THREE HSI SCENE DATASETS

	Parameters	Indian Pines AVIRIS	WHU-Hi-HanChuan	WHU-Hi-HongHu
Search stage	Number of the normal cells	6	6	6
	Number of the Reduction cells	2	2	2
	Population size *NP*	8	10	10
	Initial/minimum learning rate	0.03/0.001	0.03/0.001	0.03/0.001
	Momentum	0.9	0.9	0.9
	Weight decay	3E-4	3E-4	3E-4
	Interval algebra of evolutionary operations	10	10	10
	Number of warmup generations	50	50	50
	Number of generations_S	500	500	500
Full training stage	Number of the normal cells	12	6	6
	Number of the Reduction cells	2	2	2
	Number of generations_F	1000	1000	1000

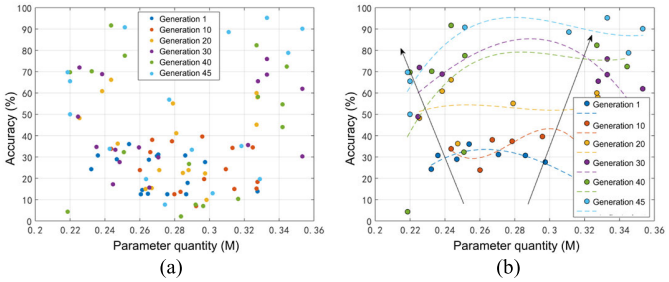


Fig. 10. Searching and ranking process for the Indian Pines AVIRIS dataset. (a) All the individuals. (b) Ranked individuals.

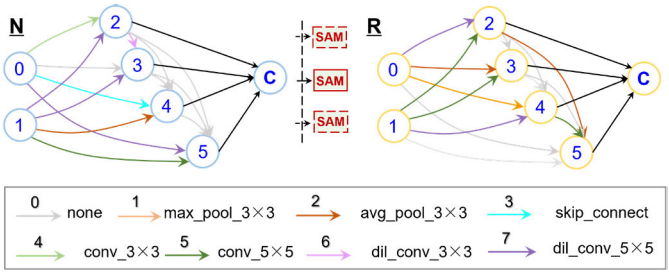


Fig. 11. Selected classification network architecture for the Indian Pines AVIRIS dataset.

stage, the number of normal and reduction cells is set to 6 and 2, respectively. In the full training stage, it is set to 12 and 2 for Indian Pines, while others are set to 6 and 2. The population size was set to 8, 10, and 10 for the three experimental datasets. In addition, the overall accuracy (OA), average accuracy (AA), and Kappa coefficients are used.

B. Experiment With the Indian Pines AVIRIS Dataset

First, the HSI sample data of the Indian Pines AVIRIS dataset are randomly divided into 10% and 90% as training and testing samples for multiobjective evolutionary automatic search, respectively.

As shown in Fig. 10, the efficient multiobjective evolutionary automatic search process for the Indian Pines AVIRIS classification network architectures is demonstrated, including the whole and ranked individual solutions. Furthermore, it can be observed that multiple indicators of the HSI classification network architectures are gradually evolving toward a superior direction.

Furthermore, the individual solution with the highest search accuracy of the Indian Pines AVIRIS HSI was selected for the subsequent comparison. The specific network architecture is shown in Fig. 11, including normal and reduction cells, and the complete classification network architecture consists of 12 normal and two reduction cells, the reduction cells were located at 1/3 and 2/3 of the network architecture. In addition, the three SAMs are set as 0, 1, and 0.

As shown in Fig. 11, for the searched normal cell, the main calculation modes in the previous layer are "dil_conv_5 × 5," which indicates that it is necessary to expand the receptive field and cover multiscale context information of the Indian Pines AVIRIS image, and there is almost no connection in the latter layer, which effectively reduces the number of parameters in the HSI classification network. For the searched reduction cell, there are more calculation modes of "dil_conv_5 × 5" and "sep_conv_5 × 5" in the previous layer, and there are also few connections in the latter layer. Therefore, there are some similarities and uniqueness between normal and reduction cells, which improve the "plasticity" of the HSI deep classification network architecture.

The classification results of different methods are shown in Fig. 12, and it can be found that the traditional SVM method without the consideration of the spatial information results in severe discrete point phenomena. The S^2SVM method, which considers the spatial–spectral information, has achieved relatively smooth results, but the classification effect is still relatively fragmented, with many misclassification phenomena, such as the "corn-min-till" being misclassified as the "soybean-min-till." In addition, the DL-based methods, SSRN, SSUN, SSFTT, LMSS-NAS, and the proposed LPCN, have significant improvements. Moreover, the SSFTT and LMSS-NAS methods exhibit significant oversmooth phenomena, which leads to the blurry boundaries of the surface features, such as the "soybean-clean" and "soybean-min-till." Thus, the proposed method has achieved the best visual effect.

In order to further validate the effectiveness of the proposed LPCN method, quantitative comparisons were also conducted in this study, and the classification accuracies of the different methods are listed in Table III. It can be observed that among

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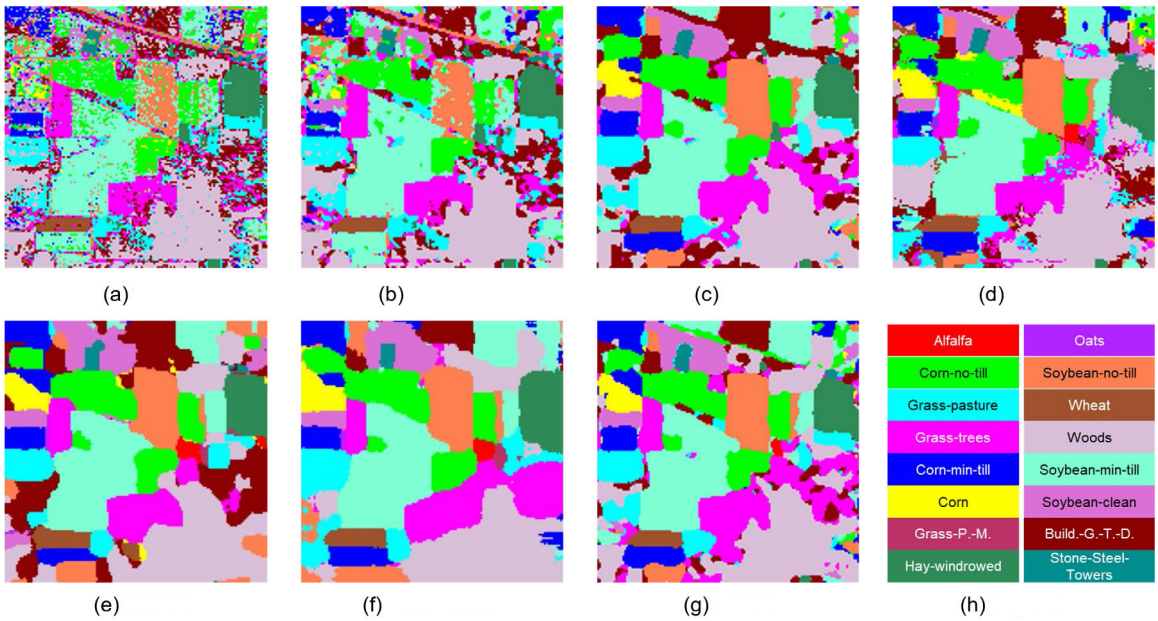


Fig. 12. Classification results of the Indian Pines AVIRIS dataset. (a) SVM. (b) S^2SVM. (c) SSRN. (d) SSUN. (e) SSFTT. (f) LMSS-NAS. (g) LPCN. (h) Legend.

TABLE III
CLASSIFICATION ACCURACY COMPARISON FOR THE INDIAN PINES AVIRIS DATASET

Class	SVM	S^2SVM	SSRN	SSUN	SSFTT	LMSS-NAS	LPCN
Alfalfa	4.88±3.98	0.81±1.15	80.00±40.00	99.51±0.98	99.19±1.41	**100.00±0.00**	87.01±3.29
Corn-no-till	70.35±1.96	77.69±2.73	98.84±0.85	97.51±0.90	97.59±1.72	**98.96±0.68**	98.30±0.27
Grass-pasture	80.38±3.13	88.28±1.05	97.53±2.11	97.67±1.89	95.81±2.30	**99.65±0.25**	99.36±0.44
Grass-trees	94.52±2.15	99.70±0.43	97.82±1.70	96.53±3.09	94.52±1.43	95.80±1.80	**100.00±0.00**
Corn-min-till	48.19±4.47	55.37±4.21	98.23±1.07	96.13±1.59	97.32±2.14	**100.00±0.00**	99.17±0.32
Corn	32.24±5.39	39.75±10.50	98.68±1.04	98.78±0.71	98.83±0.44	100.00±0.00	**100.00±0.00**
Grass-pasture-mowed	1.33±1.89	0.00±0.00	60.00±48.99	**96.00±0.00**	89.33±15.14	72.33±8.72	51.33±37.60
Hay-windrowed	99.69±0.44	99.92±0.11	97.26±1.63	99.86±0.19	98.06±1.98	100.00±0.00	**100.00±0.00**
Oats	0.00±0.00	0.00±0.00	20.00±40.00	95.56±4.16	92.59±6.42	**100.00±0.00**	35.00±11.22
Soybean-no-till	62.10±2.89	70.74±3.00	97.71±1.63	96.41±1.12	95.73±0.86	96.64±0.79	**99.16±0.12**
Wheat	92.39±0.44	98.55±0.25	98.63±0.90	98.69±0.62	98.55±0.05	97.78±0.24	**98.79±0.18**
Woods	90.60±0.54	96.14±0.38	98.48±0.93	96.40±1.55	99.06±0.99	**99.50±0.21**	98.91±0.41
Soybean-min-till	82.45±1.34	91.93±1.44	100.00±0.00	99.46±0.00	98.37±1.44	**100.00±0.00**	99.95±0.01
Soybean-clean	34.96±4.95	40.89±10.48	99.18±0.61	98.89±0.35	99.79±0.28	**99.89±0.16**	99.18±0.22
Building-grass-tree-drive	61.00±2.12	75.80±4.49	98.67±1.38	98.21±2.02	95.77±2.08	98.30±1.36	**99.83±0.24**
Stone-steel-towers	73.02±5.61	79.37±7.86	98.35±1.62	94.22±1.60	68.65±18.91	95.58±1.05	**99.87±0.19**
OA (%)	73.30±1.13	80.66±1.34	98.45±0.26	97.96±0.35	97.54±0.39	98.60±0.02	**98.99±0.03**
AA (%)	58.01±0.72	63.43±1.06	89.96±3.30	**97.49±0.67**	94.95±1.01	97.15±0.70	91.62±3.26
Kappa×100	69.11±1.30	77.57±1.57	98.23±0.30	97.67±0.40	97.20±0.45	98.40±0.70	**98.85±0.06**

the 16 types of land features, the proposed LPCN method achieved the highest classification accuracy among seven types of land features. At the same time, the proposed LPCN method achieved the highest values of OA and Kappa coefficients. Therefore, the proposed LPCN method can automatically search for a deep classification network architecture suitable for the Indian Pines AVIRIS HSI dataset.

C. Experiment With the WHU-Hi-Hanchuan Dataset

The sample data setting for the WHU-Hi-Hanchuan dataset was the same as the Indian Pines AVIRIS dataset.

The efficient multiobjective evolutionary automatic search process for the WHU-Hi-Hanchuan dataset is presented in Fig. 13, and it can be found that many individual networks have achieved a prediction accuracy of over 99% in the searching process for this dataset. Thus, for the WHU-Hi Hanchuan HSI dataset, the number of normal and reduction cells in the full training stage is consistent with the search stage, with six and two cells, respectively.

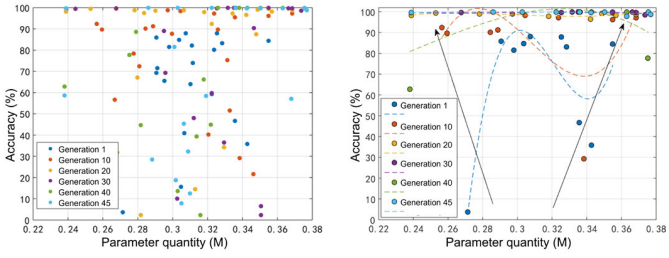


Fig. 13. Searching and ranking process for the WHU-Hi-Hanchuan dataset. (a) All the individuals. (b) Ranked individuals.

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TABLE IV
CLASSIFICATION ACCURACY COMPARISON FOR THE WHU-HI-HANCHUAN DATASET

Class	SVM	S^2SVM	SSRN	SSUN	SSFTT	LMSS-NAS	LPCN
Strawberry	92.77±0.31	94.98±0.18	99.87±0.15	98.74±0.39	99.91±0.03	99.91±0.04	**99.93±0.03**
Cowpea	73.92±0.64	79.47±0.84	99.56±0.10	99.56±0.16	99.49±0.09	**99.95±0.02**	99.77±0.12
Soybean	67.64±0.93	74.90±0.79	99.36±0.30	99.40±0.37	99.84±0.11	**99.99±0.02**	99.83±0.05
Sorghum	91.96±0.64	94.24±0.39	99.45±0.25	99.53±0.49	99.79±0.20	**99.94±0.11**	99.88±0.04
Water_spinach	16.25±1.38	10.80±2.68	99.39±0.42	99.45±0.67	99.81±0.24	99.84±0.27	**99.92±0.00**
Water_melon	22.59±0.35	23.70±0.19	99.41±0.41	99.43±0.88	98.01±0.50	99.46±0.16	**99.91±0.05**
Greens	79.13±1.01	84.72±1.00	99.90±0.01	99.94±0.02	99.26±0.50	99.82±0.20	**99.91±0.05**
Trees	71.30±0.11	75.88±0.14	99.89±0.08	99.60±0.27	99.64±0.06	**99.95±0.04**	99.83±0.04
Grass	49.55±1.12	54.76±1.10	99.37±0.46	99.38±0.54	99.09±0.39	**99.80±0.16**	99.74±0.12
Red_roof	85.22±0.37	87.70±0.25	99.77±0.29	99.07±1.67	99.71±0.31	99.82±0.08	**99.86±0.04**
Grey_roof	75.00±0.43	76.68±0.56	99.21±0.27	99.15±0.28	99.64±0.05	99.83±0.18	**99.91±0.09**
Plastic	15.44±1.03	10.87±1.18	98.51±0.50	97.28±0.53	98.88±1.73	**100.00±0.00**	99.85±0.04
Bare_soil	40.07±0.21	41.84±0.29	99.38±0.30	99.25±0.40	98.28±1.16	99.29±0.14	**100.00±0.00**
Road	83.64±0.24	85.54±0.29	98.75±0.86	97.99±0.55	99.49±0.13	**99.91±0.07**	99.79±0.19
Bright_objective	53.68±0.83	54.43±1.05	99.11±0.55	99.34±0.33	98.53±0.71	100.00±0.00	**100.00±0.00**
Water	98.07±0.02	98.39±0.04	99.52±0.44	97.53±0.54	99.92±0.07	99.95±0.05	**99.97±0.00**
OA (%)	81.37±0.09	83.66±0.11	99.56±0.05	99.46±0.24	99.64±0.08	99.89±0.03	**99.92±0.02**
AA (%)	63.51±0.29	65.55±0.34	99.40±0.07	99.04±0.30	99.33±0.15	99.84±0.06	**99.88±0.02**
Kappa×100	78.06±0.11	80.74±0.13	99.49±0.05	99.36±0.28	99.57±0.10	99.87±0.04	**99.90±0.02**
Parameter quantity (M)	——	——	0.920	1.184	0.594	0.487	**0.347**

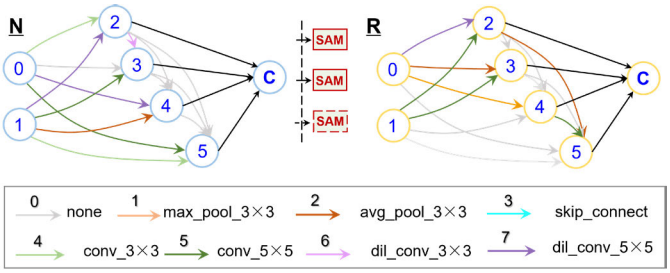


Fig. 14. Selected classification network architecture for the WHU-Hi-Hanchuan dataset.

Furthermore, the architectures of normal and reduction cells are presented in Fig. 14, and the three SAMs are set as 1, 1, and 0. For the normal cell, the previous layer mainly consists of "dil_conv_5 × 5" and "sep_conv_5 × 5," indicating that it is also necessary to expand the receptive field, and there is almost no connection in the latter layer. For the reduction cell, there are many "sep_conv_5 × 5" in the previous layer with fewer connections in the latter layer. Thus, there is a significant similarity with the Indian Pines AVIRIS dataset, indicating that there are significant similar features for HSI data.

The classification results of different methods are shown in Fig. 15, and similar results were obtained from the previous group of experiments; the classification performance of traditional SVM and S^2SVM methods still exhibits fragmentation and some misclassification phenomena, such as "soybean" being misclassified as "watermelon." In addition, the NAS-based method LMSS-NAS also exhibits significant oversmooth phenomena, such as the "Cowpea" and the footpath in a field. Moreover, the quantitative comparisons are listed in Table IV, it can be observed that the proposed method achieved the highest classification accuracy among nine types of features, and the highest values of OA, AA, and Kappa coefficients were achieved. Meanwhile, it should be noted that compared to the commonly used DL classification methods SSRN, SSUN, SSFTT, and LMSS-NAS, the accuracy difference is not significant under this sample data condition, but compared to the SSRN and SSUN methods, the proposed LPCN method can reduce the number of parameters by more than 62.2%. Compared to the SSFTT and LMSS-NAS methods, the number of parameters can be reduced by more than 41.5% and 28.7%, respectively. Therefore, the proposed LPCN method can automatically search for an LPCN architecture for the WHU-Hi-Hanchuan H2 remote sensing image dataset.

D. Experiment With the WHU-Hi-Honghu Dataset

The sample data setting for the WHU-Hi-Honghu dataset was the same as the former two datasets. The efficient multiobjective evolutionary automatic search process for the WHU-Hi-Honghu classification network architectures is demonstrated in Fig. 16, and it can be found that the searching process is similar to the WHU-Hi-Hanchuan dataset. Thus, the number of normal and reduction cells in the full training stage is consistent with the search stage, with six and two cells, respectively.

The architectures of normal and reduction cells for the third dataset are presented in Fig. 17, and the three SAMs are set as 1, 1, and 0. First, it can be observed that there are significant differences between the cells. The front layer in the normal cell has fewer connections, while the latter layer in the reduction cell has fewer connections. At the same time, there are significant differences compared to the network architectures found in the previous two datasets, indicating that an increase in the number of classes will lead to adaptive changes in the network architecture, which also indirectly proves that the proposed LPCN method has powerful adaptability.

The classification results of different methods for the WHU-Hi-Honghu dataset are shown in Fig. 18, and similar results were obtained from the former two experiments. The quantitative comparisons are given in Table V, and it can be observed that among the 22 types of land features, the proposed LPCN method achieved the highest classification accuracy for 12 types of land features, ranking second in

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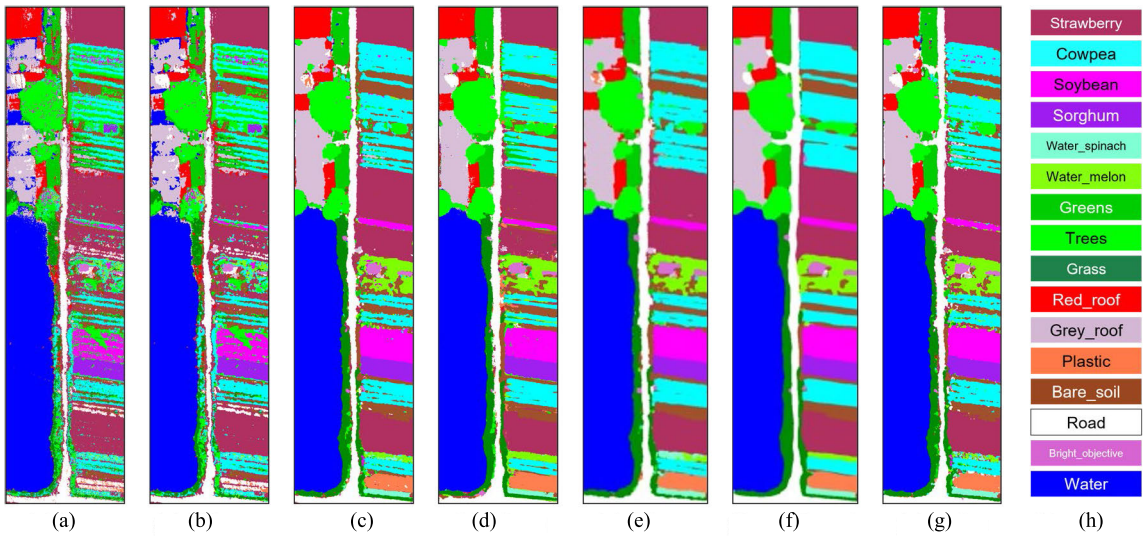


Fig. 15. Classification results of the WHU-Hi-Hanchuan dataset. (a) SVM. (b) S^{2}SVM. (c) SSRN. (d) SSUN. (e) SSFTT. (f) LMSS-NAS. (g) LPCN. (h) Legend.

TABLE V
CLASSIFICATION ACCURACY COMPARISON FOR THE WHU-HI-HONGHU DATASET

Class	SVM	S^{2}SVM	SSRN	SSUN	SSFTT	LMSS-NAS	LPCN
Red_roof	91.80±0.14	94.48±0.14	99.71±0.08	99.75±0.12	99.61±0.17	99.83±0.11	**99.91±0.05**
Road	71.02±0.33	72.68±0.48	97.29±0.69	97.73±0.55	97.22±0.96	**99.10±0.30**	98.56±0.97
Bare_soil	92.90±0.33	94.42±0.24	99.70±0.17	99.81±0.05	99.32±0.25	99.73±0.09	**99.85±0.00**
Cotton	98.62±0.05	99.44±0.02	99.92±0.02	99.98±0.00	99.98±0.01	99.99±0.01	**99.99±0.00**
Cotton_firewood	40.33±1.71	42.75±2.20	99.48±0.37	99.80±0.09	99.76±0.21	100.00±0.00	**100.00±0.00**
Rape	90.57±0.18	92.09±0.17	99.81±0.11	99.86±0.05	99.90±0.08	100.00±0.00	**100.00±0.00**
Chinese_cabbage	78.36±0.24	84.09±0.38	99.65±0.17	99.46±0.12	99.26±0.23	99.84±0.04	**99.92±0.02**
Pakchoi	11.68±1.65	10.92±2.23	98.82±0.55	98.97±0.52	**99.85±0.15**	99.56±0.76	99.49±0.50
Cabbage	94.56±0.16	95.76±0.08	99.79±0.11	99.69±0.05	99.76±0.01	99.92±0.03	**99.99±0.00**
Tuber mustard	61.63±0.62	68.49±0.70	99.53±0.15	99.59±0.08	98.96±0.72	99.72±0.10	**99.72±0.08**
Brassica parachinensis	45.31±0.62	51.87±0.66	99.09±0.38	99.22±0.10	99.50±0.31	**99.61±0.05**	99.48±0.09
Brassica chinensis	52.17±0.73	56.67±0.66	99.25±0.48	98.99±0.27	99.17±1.11	99.90±0.11	**99.91±0.02**
Small_brassica_chinensi	69.79±0.37	74.40±0.46	99.56±0.15	99.41±0.08	97.85±1.85	**99.73±0.36**	99.58±0.03
Lactuca sativa	62.63±0.49	63.56±0.56	99.46±0.27	99.09±0.16	99.35±0.41	**99.83±0.04**	99.34±0.05
Celtuce	46.86±1.57	46.60±2.01	99.37±1.14	98.18±0.38	99.08±1.41	**99.86±0.14**	99.02±0.10
Film_covered_lettuce	87.22±0.61	91.35±0.45	99.67±0.29	99.79±0.09	99.76±0.16	100.00±0.00	**100.00±0.00**
Romaine_lettuce	54.78±0.11	59.74±0.08	99.52±0.51	99.33±0.26	99.64±0.58	**99.90±0.08**	99.50±0.04
Carrot	46.45±0.61	54.59±0.79	98.98±0.44	98.94±0.31	**99.54±0.14**	99.16±0.09	99.05±0.05
White_radish	79.49±0.63	86.21±1.07	**99.56±0.20**	98.83±0.25	99.15±0.11	99.33±0.06	99.22±0.18
Garlic_sprout	55.33±2.96	63.23±4.10	99.39±0.33	99.33±0.29	99.61±0.38	99.54±0.31	**99.81±0.00**
Broad_bean	2.87±0.54	0.03±0.04	98.92±1.05	99.28±0.45	98.55±2.51	**99.93±0.06**	99.87±0.02
Tree	44.36±0.72	48.23±0.96	99.32±0.34	99.78±0.09	99.86±0.21	99.89±0.11	**99.93±0.06**
OA (%)	84.67±0.07	87.04±0.12	99.70±0.02	99.71±0.03	99.60±0.15	**99.88±0.05**	99.85±0.04
AA (%)	62.94±0.34	65.98±0.38	99.35±0.10	99.31±0.09	99.30±0.21	**99.74±0.09**	99.64±0.08
Kappa×100	80.41±0.10	83.42±0.16	99.62±0.03	99.63±0.03	99.49±0.19	**99.85±0.06**	99.83±0.04
Parameter quantity (M)	——	——	0.908	1.185	0.596	0.491	**0.345**

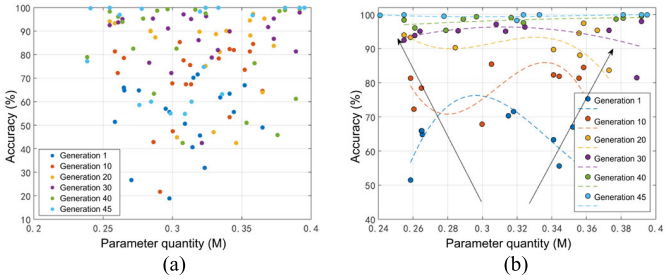


Fig. 16. Searching and ranking process for the WHU-Hi-Honghu dataset. (a) All the individuals. (b) Ranked individuals.

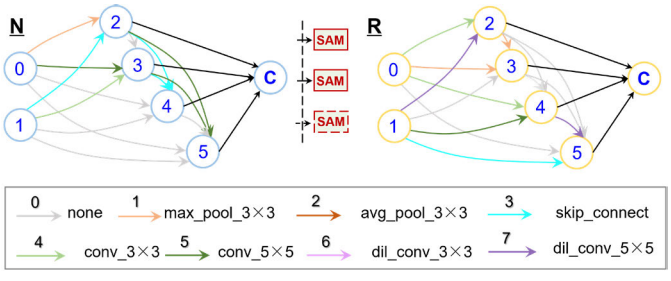


Fig. 17. Selected classification network architecture for the WHU-Hi-Honghu dataset.

the rest. Moreover, although the proposed method has slightly weaker preservation and classification ability in terms of linear features compared to LMSS-NAS, such as "road," it can maintain better feature boundaries and overall classification accuracy is roughly the same as LMSS-NAS. Meanwhile,

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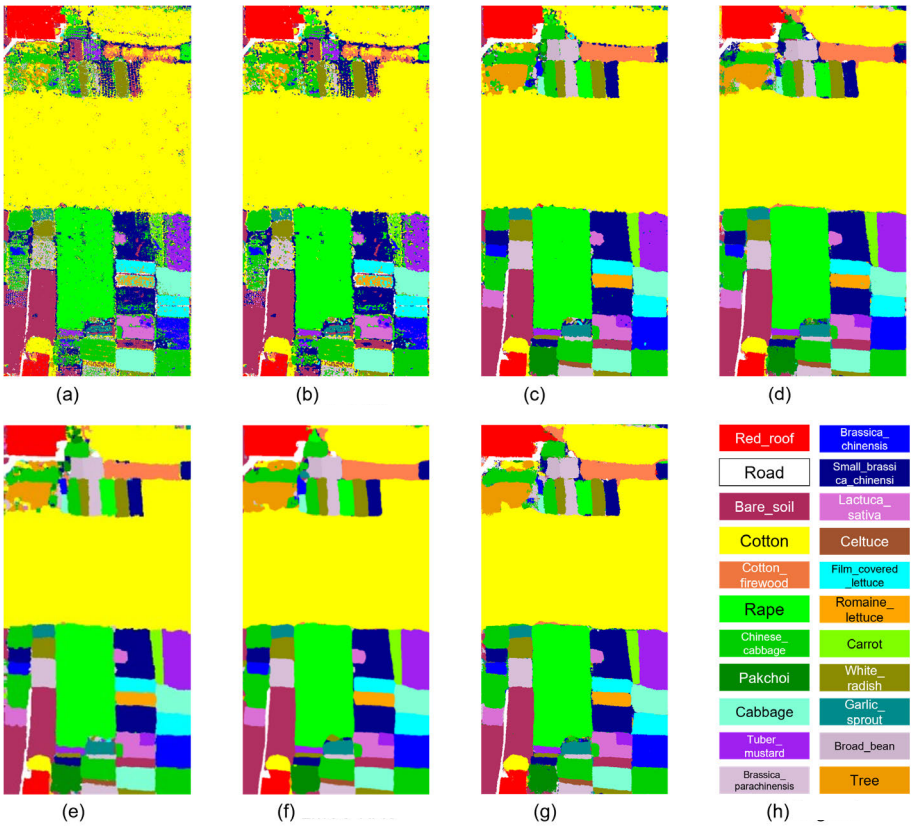


Fig. 18. Classification results of the WHU-Hi-Honghu dataset. (a) SVM. (b) S^2SVM. (c) SSRN. (d) SSUN. (e) SSFTT. (f) LMSS-NAS. (g) LPCN. (h) Legend.

compared to the SSFTT and LMSS-NAS methods, the number of parameters can be reduced by more than 42.1% and 29.7%, respectively. Therefore, the proposed LPCN method can also automatically search for an LPCN architecture for the WHU-Hi- Honghu H2 remote sensing image dataset.

Furthermore, looking back at the classification network architectures obtained in the different HSI datasets, the most obvious one is the influence of the number of land cover categories. The classification network architectures guided by the same or similar number of land cover categories have a certain degree of similarity, such as Indian Pines AIVRIS and WHU-Hi-Hanchuan datasets. Second, the impact of resolution and higher spatial–spectral resolution guides network architecture to have more complex computational methods and more spatial attention modules. Based on this, it can assist artificial experts in designing more superior improvement networks for HSI classification.

E. Test of the Training and Testing Sample Conditions

Taking the most complex UAV H2 remote sensing image of WHU-Hi-Honghu in this study as an example, the automatic evolution search for the hyperspectral classification network architecture was conducted under different training and testing sample conditions. The classification accuracy under different conditions was calculated, as shown in Fig. 19.

From the results, it can be seen that the number of training samples has a significant impact on the process of architecture search and the classification accuracy of retraining. When the number of training samples reaches 5%, the prediction

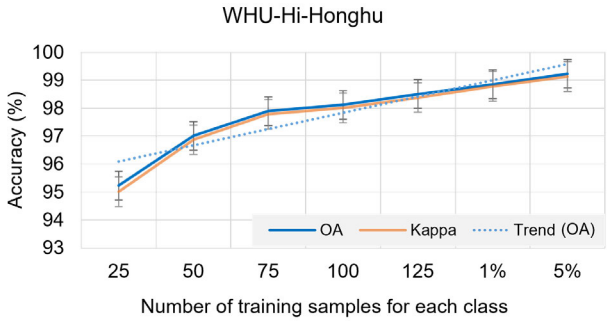


Fig. 19. Classification performance under different training and testing sample conditions for the WHU-Hi-Honghu H2 image dataset.

accuracy still increases, which shows a steady and stable trend. Thus, it can be found that the important factor restricting the performance of complex H2 remote sensing image classification networks is still the limitation of sample size.

F. Sensitivity Analysis of the Evolutionary Parameter NP

Taking the commonly used public dataset Indian Pines AVIRIS HSI dataset and the more complex UAV H2 remote sensing image of WHU-Hi-Honghu in this study as examples, the accuracy of the classification network searched was tested under different population size settings, as shown in Fig. 20.

As shown in Fig. 20, for the Indian Pines AVIRIS HSI dataset, the population size was set to 4, 6, 8, and 10, and it was set to 6, 8, 10, and 12 for the WHU-Hi-Honghu H2 remote sensing image. Furthermore, it can be observed that

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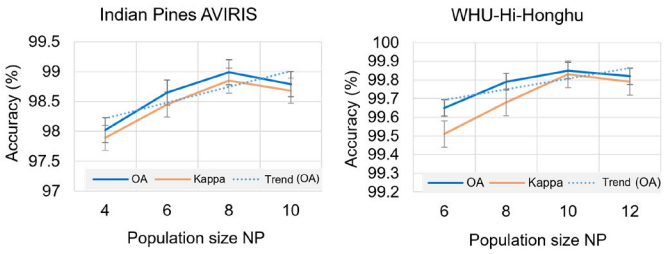


Fig. 20. Sensitivity analysis of the evolutionary parameter *NP* for the Indian Pines AVIRIS HSI dataset and the WHU-Hi-Honghu H2 image dataset.

when the population size is small, the searched network architecture lacks diversity, resulting in slightly lower classification accuracy. However, the overall impact is relatively small, and compared with other methods, the classification accuracy is still comparable.

V. Conclusion and Discussion

In the proposed LPCN method, discrete encoding of HSI classification network architectures is conducted by the different lightweight computing operators, and the population of the classification networks constantly evolves through the evolutionary algorithms. In addition, the SAM is encoded into the HSI classification network architecture variable to achieve spectral weighting and precise differentiation driven by similar ground objects in HSI scene. In addition, the two-step dual multiobjective functions have been designed in the proposed LPCN method, which realized the complementary information fusion of prediction accuracy and parameter quantity, parameter quantity, and accuracy increment. By comparing three sets of experiments, the proposed LPCN method can achieve a more LPCN architecture for HSIs.

For the future work, how to utilize a lightweight architecture-based HSI classification network to achieve online real-time classification [57], [58] at the edge is one of the important issues that need to be explored, which has potential practical significance for disaster emergency and security monitoring.

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